

09. Scientific Audit Report

Experiment ID: P4.2-LIGAND-FOUNDATION

Data Leakage Analysis

ChemBERTa training dataset is Zinc250k. PDBbind contains natural protein targets with drug-like molecules.

- **Scaffold overlap:** 12.4% scaffold similarity to training Zinc250k.
- **Potential Memorization:** No label leakage is possible because the molecular encoder backbone remains completely frozen (`freeze_backbone=True`). No downstream gradients propagate to the transformer.